structures shall be incorporated into and maintained for the duration of the project along all detour, access and haul roads to minimize erosion and maintain drainage patterns.

165.09.03 Abandonment - The Contractor shall winterize haul and access roads at the end of the construction season and shall ensure proper drainage control measures are in place. Culvert inlets and outlets shall be flagged. During periods of work stoppages, the Contractor shall ensure sufficient inspection and maintenance of the access, detour and/or haul roads such that adverse environmental impacts are avoided or minimized. In the event that potential or existing environmental problems are identified by the Contractor, its staff, the Ministry, the Environmental Agencies or the public, the Contractor will take immediate steps to rectify the problem and ensure corrective action is taken to prevent future recurrence of similar problems.

When no longer required by the Contractor, construction roads shall be properly decommissioned, including measures such as blocking such roads off from vehicular traffic, scarifying the compacted surfaces and generally revegetating the area. All drainage and crossing structures shall be removed, and sufficient cross ditches and ditch blocks constructed and stabilized to restore the original drainage patterns and prevent erosion and may include, but may not be limited to decompaction of the road surface, revegetation of exposed soil surfaces, and reforestation.

165.10 Instream Works - In general the Contractor shall comply with the requirements specified in Section 5, "Guidelines for Construction Practices within the Fisheries Sensitive Zone", of the “Land Development Guidelines”, as well as those requirements noted below. The Contractor, , shall prepare a detailed set of environmental procedures for any work within a Fisheries Sensitive Zone, within the wetted perimeter of any fish-bearing watercourse (or any watercourse that flows directly into a fish-bearing stream), or for any work with the potential to cause major adverse impacts on a fish-bearing watercourse (e.g., concrete pour for a bridge deck over a fish-bearing watercourse, hydroblasting of bridge substructures over a fish-bearing watercourse, etc.).

No clearing or equipment operation shall take place in a Fisheries Sensitive Zone prior to Environmental Agency approvals. Waste material generated during instream works shall not be stored or dumped within the floodplain unless otherwise approved.

Drainage of waterbodies, such as wetlands, swamps or beaver ponds, shall be subject to the approval of the Environmental Agencies, unless drainage of these areas is specifically required as part of the Contract and has been approved by Fisheries and Oceans Canada and British Columbia Water Management office.

165.10.01 Provision of Fish Passage - Unless the Contractor has written approval from the Environmental Agencies, any instream works in fish bearing streams shall provide for fish passage. It is the Contractor's responsibility to contact the Environmental Agencies to determine fish passage requirements for the specific location.

The Contractor is required to give two week's notice to the Environmental Agencies and the Ministry Representative of any activities, which may result in stranding of fish, to allow for scheduling of fish retrieval and/or transfer operations.

165.10.02 Management of Water Discharges - During preparation of stream diversions, culvert installations and other operations involving dewatering where drainage could readily reach a designated watercourse, all effluent and silt-laden water shall be discharged to a sediment control pond or a vegetated area for removal of silt prior to its release into that watercourse. This requirement also applies to the control of discharge resulting from curing areas of recent concrete pours for bridges or drainage structure headwalls. The direct discharge or discharge via seepage of untreated, silt-laden water or other deleterious substance into any watercourse is prohibited.

165.10.03 Encroachment of Fill into Watercourses - Embankments which encroach on fish-bearing watercourses shall either be isolated and dewatered, by means of a cofferdam, until surfaced with riprap, or constructed of clean granular material and riprap or shot rock which is free of fine-grained material or other potential contaminants. Riprap composition and placement may be further detailed in the Contract. The method of placement of this material shall be such that disturbance and/or alteration of aquatic habitats is absolutely minimized. Fill material must be entirely contained within the proposed fill cross section.

165.10.04 Temporary Stream Crossings - Prior to construction, the Contractor shall provide the Ministry Representative with a plan describing the proposed locations and types of stream crossings, complete with construction procedures and timing of construction. The Contractor shall forward a copy of the approved plan to the Ministry Representative. Temporary stream crossings shall be subject to the same environmental constraints as permanent crossings, and shall be built to pass, at least, the 10 year return period flood for the time of year during which the structure will be in place. Wherever possible, vegetative cover shall be maintained in order to minimize erosion.

Temporary stream crossings that have been constructed during periods of low precipitation (i.e. summer and late winter) shall be completely removed prior to periods of

increased precipitation (i.e. fall and spring freshet), unless otherwise approved by the Environmental Agencies.

165.10.05 Instream Equipment Operation or Crossings - The entering or crossing of any designated watercourse by construction equipment, when not specified in the Contract, shall require and comply with the written approval of the Environmental Agencies. The Contractor shall notify the Ministry Representative and the Environmental Agencies 48 hours prior to allowing equipment to cross or enter designated streams. The Contractor's Environmental Monitor shall be present while equipment enters into and exits from any designated watercourse.

Upon completion, the banks shall be restored, stabilized and revegetated to prevent erosion. This work shall be completed to the satisfaction of the Environmental Agencies.

165.10.06 Culvert Installations - Culvert installation in designated watercourses shall involve an appropriate method of isolating the work site from the stream, such as the diversion of the stream around the culvert site, and the placement of the culvert in the "dry".

Installation of riprap and other protective works shall be carried out at the earliest possible time following culvert installation in order to prevent erosion and siltation.

165.10.07 Culvert Obstructions - Unless the watercourse is dry, an obstructed culvert shall be cleared by mechanical means. Any variance to using mechanical means to clear wet obstructed culverts must have the prior approval of the Workers' Compensation Board.

165.10.08 Channel Diversions - Channel diversions shall be performed in accordance with the Contract. If not specified in the Contract, methods shall be approved by the Environmental Agencies, and shall comply with the "Land Development Guidelines".

165.10.09 Bridge Construction - Bridge Endfills shall be constructed implementing fully the applicable subsections of DBSS 165, and DBSS 769, Protection and Retention of Vegetation. The Fisheries Sensitive Zone shall not be disturbed until construction of the endfill or crossing structure is ready to proceed. The Contractor may be required to hand fall trees and leave the root systems in place.

165.10.10 Watercourse Clean-up - Final watercourse clean-up shall consist of, but not be restricted to, the removal of temporary haul roads and temporary crossings, the reshaping of the stream to its original configuration, width and depth, the protection of stream banks, revegetation, and the removal of all construction related material and debris, including any material and debris deposited upstream or downstream of the site as a result of operations All abandoned temporary diversions shall be plugged and stabilized. In the event that erosion has occurred during the course of construction, the Contractor will be required to recontour, stabilize and revegetate the affected area.

165.11 Ditch Maintenance - To reduce the risk of damage to aquatic habitat and sensitive life stages of fish and other aquatic organisms, the Contractor shall ensure that the excavation and maintenance of ditches is conducted in an environmentally sound manner.

Excavation for ditch cleaning of any ditch that contains or directly drains into fish-bearing waters shall be performed only within Timing Windows as detailed in the "Guidelines for Land Development, Appendix III" or as specified in the Contract.

The Contractor shall consult local Environmental Agencies for the locations of sensitive areas and local Timing Windows prior to commencement of ditch maintenance. Areas identified as Fisheries Sensitive Zones shall be revegetated by the Contractor prior to the close of the timing window.

165.12 Bridge Maintenance - In performing any bridge rehabilitation or maintenance, the Contractor shall follow the procedures specified in the document "Federal Fisheries Guidelines for the Protection of Fish and Fish Habitat during Bridge Maintenance Operations in British Columbia"

The Contractor shall take measures to ensure absolute minimal loss of paint and abrasive material into watercourses during bridge cleaning. Mitigation measures may include placement of ground covers, lined nets or tarpaulins to capture falling debris, blast enclosures to encase the section of the bridge to be abrasively blasted, use of vacuum-shrouded power tools, and use of floating booms and barges to trap floating debris. Similar measures shall also be employed to assist in mitigating paint spray drift.

165.13 Servicing and Operation of Equipment

165.13.01 Transport of Materials - The Contractor shall use equipment and containers that are capable of safely transporting petroleum products and/or hazardous materials in compliance with Section 7.33.1 of the Federal "Transport of Dangerous Goods Regulations" for bulk containers, and Sections 7.21 and 7.23 of the same regulations for materials in packages or small containers. These regulations shall apply to both on-road and on-site transport. In addition to compliance with the Federal Transport of Dangerous Goods Regulations, the

transportation of fuel trailers, fuel tanks on skids, and non-commercial fuel trucks shall comply with all the requirements of the Motor Vehicle Act and the British Columbia Fire Code.

The Contractor shall take the necessary precautions to prevent the loss of materials during transport on public highways, roads, access roads, and haul routes. Trucks carrying loose materials shall be covered and restraints shall be used to prevent materials from blowing or falling from vehicles. The Contractor shall be responsible for collection and removal of any and all litter deposited by vehicles or equipment along access routes during construction-related activities, including removal of dirt and mud deposited from truck tires on municipal roadways.

165.13.02 Location and Installation of Fuel Storage Facilities - The siting and installation of all fuel storage facilities shall be the responsibility of the Contractor. Fuel storage shall not be located within 30 m of a watercourse, within a watercourses flood plain, or where there is a potential for any spilled fuel to enter a watercourse or groundwater. Fuel storage facilities shall be located on flat or gently sloping ground and shall be dyked to contain at least 125% of the total capacity of the storage containers. Dykes shall be constructed of impermeable material or lined to ensure that petroleum products cannot escape.

All large fuel storage tanks must be locked and secured when not in use. Automatic shut-off nozzles shall be installed on all dispensing units over 250 ℓ capacity. Fuel storage tanks must be drained within one week of completion of construction or within one month of a prolonged shutdown period.

All small fuel storage containers, such as 45 gallon (200 litre) drums, used as a fuel cache shall be installed on a stable storage rack, within an impermeable containment device capable of capturing at least 125% of the total capacity of the storage container(s). A cover, such as a tarp, must be placed over the top of the fuel cache to prevent accumulation of precipitation in the containment device. The small fuel storage container(s) shall contain a metal spigot with a padlock placed on the container when not in use.

In the transport, storage and/or dispensing of fuel and other petroleum-based products, the Contractor shall comply with the requirements outlined in the Ministry of Environment publication: "Environmental Standards & Guidelines for Fuel Handling, Transportation and Storage" (December, 1995).

The Contractor shall be responsible for all costs associated with clean-up and disposal of any escaped toxic and hazardous substances.

165.13.03 Servicing of Equipment - The fueling, servicing or washing of machines or equipment within the wetted perimeter or riparian zone of watercourses is prohibited. On-site fueling and lubrication of equipment shall also be conducted as far as possible from detention and sediment control facilities. Servicing should be conducted at designated sites that are properly protected and approved by Fisheries and Oceans Canada and British Columbia Ministry of Environment.

Refuse generated during the servicing of equipment (e.g., air and oil filters, hydraulic fluids, petroleum products) shall be collected and disposed of in an environmentally acceptable location and manner. Where possible, the Contractor is encouraged to recycle lubricants and other waste materials generated during the servicing of equipment and machinery, or alternatively dispose of such materials and refuse at nearby recycling depots. The dumping of oil or other deleterious materials on the ground or in any watercourse is strictly prohibited. The Contractor shall provide a means of catching and retaining drained oil or other deleterious materials and shall properly dispose of these materials .

165.13.04 Equipment Operation in Environmentally Sensitive Areas - Construction equipment shall be operated only within the designated construction site and access roads. Equipment operators shall not be allowed to damage or destroy vegetation or streambanks outside of this area.

The operation of equipment and machinery in a Fisheries Sensitive Zone must be:

a**)** authorized by the Environmental Agencies

b) kept to an absolute minimum, and

c) undertaken only during the period(s) specified in the Contract.

When working in any designated watercourse, equipment shall use biodegradable hydraulic fluid and shall be steam cleaned of oil, grease and other contaminants deleterious to aquatic species, prior to commencing work. When working in or near any watercourse, the Contractor shall ensure that all hydraulic systems, fuel systems and lubricating systems are in good repair. Equipment with fuel or fluid leaks shall not be permitted to enter the wetted perimeter of any watercourse. Equipment developing such leaks shall be removed immediately and repaired.

165.14 Waste Disposal and Toxic/Hazardous Materials

165.14.01 General - All non-toxic or non-hazardous

wastes which are not designated as combustible waste to be burned on-site shall be either recycled or disposed of in an approved sanitary landfill or other specialized area as indicated in the Contract. Any waste material that is inadvertently dumped in or adjacent to watercourses or other designated environmentally sensitive areas shall be removed by the Contractor and disposed of in an approved manner at the Contractor's expense.

The Contractor shall be responsible for the regular collection and disposal of all waste material generated by employees and sub-contractors. The Contractor shall take the necessary precautions to prevent loss of these materials during transport on public highways and roads, and shall be responsible for cleanup of all of these materials and all litter deposited by employees and subcontractors along access routes during construction-related activities, at no expense to the Ministry. Construction debris shall not be allowed to accumulate on the construction site but shall be collected promptly and regularly removed from the site. The Contractor shall ensure that waste materials are placed and stored in suitable containers. Waste material shall be disposed of at an approved waste disposal site. The Contractor shall be responsible for providing and regularly servicing animal-proof refuse containers and for ensuring that all employees and subcontractors dispose of wastes to keep the Project clear of waste and garbage.

165.14.02 Spill Clean-up Plans for Toxic/Hazardous Materials - Contingency plans for the clean-up of toxic or hazardous spills shall be prepared prior to construction and submitted, together with a list of spill abatement equipment to be stored on the job site,. The Contractor shall have on site any additional equipment or materials deemed necessary to deal with a potential spill. For operations adjacent to a watercourse, an approved spill kit is to be present. The Contractor shall promptly replace any used spill abatement and clean-up materials and maintain a sufficient inventory of materials throughout construction operations. For operations being conducted adjacent to or within the wetted perimeter of a watercourse or other Environmentally Sensitive Area, the Contractor shall have an approved spill kit ready for use nearby.

The Contractor shall immediately report any spill of any toxic or hazardous material verbally to the Ministry Representative and the Provincial Emergency Program (24 hour phone line: 1-800-663-3456). Written notification of the spill must follow within two weeks of this verbal report. The Contractor shall immediately take the necessary steps to abate the discharge and provide the necessary labour, equipment, materials and absorbents to contain and remove the spill, clean up the affected area, dispose of waste materials at an approved disposal site, and restore the area to the satisfaction of the Environmental Agencies, at the Contractor's expense. Any soil contaminated by spills shall be removed and replaced by comparable substitutes at the expense of the Contractor. Contaminated soil and vegetation removed shall be disposed of in an approved waste disposal site.

The Contractor shall document any spills observed in the vicinity of the Project that are not the result of Project-related activities, and notify the Environmental Agencies of such incidents.

165.14.03 Contaminated Sites - If an area within or adjacent to the project site has been identified as a possible source of contaminated or hazardous material and the defined contaminated or hazardous material has not been removed prior to the start of the project, the Contractor shall notify the Ministry Representative at least two weeks prior to the time work is scheduled in or adjacent to that area.

In the event that the Contractor locates material on the project site believed to be contaminated or hazardous, and which has not been previously identified, the Contractor shall immediately cease work in that area and notify the Ministry Representative. The Ministry Representative will seek investigation and disposal recommendations from regional staff of the British Columbia Ministry of Environment, Environmental Protection office. No work shall proceed in the identified or suspected area until such time as these materials have been removed to the satisfaction of the Ministry Representative, the Ministry of Environment and the British Columbia Workers' Compensation Board. Unless it is specified in the Contract, or agreement is made to carry out such work, the Contractor is not responsible for the removal of contaminated or hazardous materials, which are not the result of the Contractor's actions.

165.14.04 Concrete Wastes - The Contractor shall isolate fresh concrete or cement from any designated watercourse for 48 hours after placement. Containers or trucks carrying cement or fresh concrete shall be washed at a site approved by the Environmental Agencies. Concrete wastes, including wastewater from batching or cleaning, shall only be disposed of at approved and designated disposal sites (i.e., location designated with proper signing). All cement-contaminated wastewater from cleaning or mixing is to be considered toxic, and must be prevented from entering any designated watercourse for at least 48 hours to allow the water to reach neutral pH. Any cement or concrete-contaminated wastewater shall be tested prior to release into a designated watercourse. In the event that such wastewater pH levels are unacceptable in terms of water quality for fish and other aquatic species, the Contractor shall take the necessary measures to contain and treat such wastewater until acceptable pH levels are achieved (i.e., pH levels are the same as the receiving waters).

The Contractor shall securely store on-site and use suitable

equipment and materials for the mitigation of concrete spills into or in areas adjacent to watercourses. For example, cylinders of gaseous carbon dioxide shall be kept on the Project site, which shall be used in the event of concrete or concrete leachate discharges into any watercourse. During any concrete pour within 15 m of, or in work areas above the wetted perimeter of any designated watercourse, the Contractor shall ensure that carbon dioxide cylinders and suitable application devices (e.g., weighted soaker hoses) are available on-site and ready for use in the carbonation of water columns to neutralize any concrete leachate that is inadvertently discharged into the watercourse. Where concrete leachate is allowed to enter settling ponds, such that pH levels become very high, bubbling with carbon dioxide may not be an effective buffering agent. In such cases, it may be necessary to use other methods, such as acid buffers.

Aged, broken concrete shall only be used as riprap with the approval of the Environmental Agencies. The Contractor shall remove broken concrete or concrete wastes that have been inadvertently placed in non-approved sites on the Project and dispose of them at an approved waste disposal site.

165.14.05 Petroleum Wastes - Refuse generated during the servicing of equipment shall be removed from the site and disposed of in a location and manner that has been approved. The dumping of oil or other deleterious materials (such as diesel used to wash out dump boxes) on the ground is prohibited. Where facilities are located within a reasonable distance, the Contractor shall dispose of wastes at recycling depots.

Solids, sludges and other pollutants generated as a result of construction or removed during the course of treatment or control of wastewaters shall be disposed of in a manner that prevents their direct or indirect discharge to any watercourse or groundwaters. Effluent generated by the water scrubber of an asphalt mixing plant must be given retention time in suitably sized, impermeable settling ponds prior to release into an adjacent watercourse. The required retention time will be determined by in consultation with representatives from British Columbia Ministry of the Environment.

Asphalt pavement shall be stockpiled for recycling in a specified location, or disposed of in a location approved by the British Columbia Environmental Protection branch,.

165.15 Pesticides - The use of all pesticides is subject to pre-approval. The use of herbicides for vegetation control, other than noxious weeds, is not permitted. The Contractor shall comply with all Federal, Provincial and local regulations relative to the storage, use, and proper disposal of pesticides. Relevant legislation includes the Federal Pest Control Products Act and the British Columbia Pesticide Control Act. All pesticide wastes, washwaters, solvents, and containers shall be disposed of in compliance with the British Columbia Waste Management Act, Special Waste Regulation.

165.16 Air and Noise Pollution

165.16.01 Noise and Emissions - All activities, equipment, processes and work operated or performed by the Contractor in accomplishing the specified construction shall be in strict accordance with Federal, Provincial and local regulations governing noise levels and air emission standards.

165.16.02 Dust Control - Application and handling of any dust palliative, with the exception of water, shall be in compliance with the Ministry standards as given in "Maintenance Guidelines for Dust Palliatives and Gravel Road Stabilization".

Dust control techniques may also be required during other construction operations, including but not limited to demolition, drilling, sand blasting and concrete cutting.

165.17 Use of Water

165.17.01 Authorization for Use - For any source of water not specified in the Contract, the Contractor is required to contact British Columbia Ministry of Sustainable Resource Management / Ministry of Environment Water Management office before any water is diverted, impounded, pumped or used for any purpose, including dust control, compaction, or operation of a work camp. Authorization in the form of an approval for short term use of water may be required from that agency. The Contractor should be aware that, if required, a water application approval may take at least six weeks to process.

Downstream water flow must be maintained at all times during water takings in fish-bearing waters.

Prior to commencing any work which may affect potable water supplies downstream of the construction area, the Contractor shall ensure that all owners of licensed water intakes have been notified.

Water sources must be approved. In order to reduce the impact on local water supplies and watercourses, the Contractor may be required to use uncontaminated wastewater, such as that pumped from the surface of sedimentation basins.

165.17.02 Screened Intake Requirements - The intakes of all pumps or diversions used to withdraw water from fish-bearing watercourses shall be screened in accordance with the Fisheries and Oceans Canada "Freshwater Intake

End-of-Pipe Fish Screen Guideline".

The Contractor shall provide and maintain stable access routes to all water withdrawal sites. The location of all such routes shall be subject to the approval of the Environmental Agencies.

165.18 Support Facilities

165.18.01 Location and Operation of Support Facilities - The preservation of trees, shrubs, ground cover, fish and wildlife must be considered in site selection and construction of all temporary support facilities. During site preparation, vegetation shall be removed using selective hand clearing in preference to blanket clearing with heavy machinery, retaining vegetative cover wherever possible. The Contractor shall locate temporary field offices, storage, plant and other facilities on pre-existing cleared portions of the job-site or on areas to be cleared during the course of routine construction.

Where possible, for aesthetic purposes, work camps and parking areas shall be located behind a vegetative screen buffer. Work vehicles and heavy equipment shall be parked in designated locations only.

The Contractor shall abide by all pollution control practices, laws, ordinances and regulations applicable to the construction of work camps and support facilities. The Contractor shall obtain the approval of the British Columbia Ministry of Health, the British Columbia Water, Land and Air Protection Environmental Protection and Water Management offices, and any applicable local authority prior to the installation and/or operation of sewage disposal and potable water distribution systems.

Work camps, field office facilities and work sites shall be kept clear of litter and garbage. The Contractor shall be responsible for providing and regularly servicing animal-proof refuse containers and for instructing all employees to dispose of food wastes accordingly. Sanitary landfill operations shall be located well away from the main camp in order to minimize conflicts with wildlife.

Combustible solid wastes shall be incinerated on a regular basis, subject to all applicable emission guidelines. Solid waste containers shall be large enough to contain all of the wastes generated between collection periods. The containers shall be constructed such that spilled liquids are contained and access by insects and wildlife is prevented. Storage areas and containers shall be maintained in a sanitary condition and shall be covered to prevent spreading of wastes by water, wind or animals. Transportation of solid waste shall be performed in a manner which prevents littering during transit to the disposal site.

Sanitary landfill sites shall be located a minimum of 450 m from watercourses or campsites, and shall be situated such that contamination of any stream, lake or groundwater system is avoided. The location of all sanitary landfill sites must be approved by British Columbia Ministry of Environment, Municipal or local officials. The bottom of the landfill site shall be located at least 2 m above the water table. Trees and other vegetation shall be removed progressively as the size of the landfill increases in order to minimize erosion by wind and water.

Sanitary wastes shall not be discharged into watercourses or on the soil surface. All temporary toilets shall be equipped with approved septic tanks having safe drainage or with closed holding tanks which are emptied only into approved treatment plants or sewage tanker trucks. All temporary toilets used on -site shall be placed in environmentally acceptable areas, and shall be secured to avoid or minimize damage from vandalism.

165.18.02 Abandonment of Sites - Upon completion of construction, all temporary support facilities and camp infrastructure, including buildings, equipment, lumber, refuse, surplus materials, fencing and other such items shall be removed. The original drainage pattern shall be re-established, and all disturbed areas shall be revegetated.

The Contractor shall ensure that all project-related sites are left in an environmentally acceptable manner, subject to the approval of the Ministry Representative and the Environmental Agencies. Decommissioned construction areas, material sources and access roads shall be reclaimed by the Contractor such that as much of the original forest, wildlife, agricultural productivity, recreational use, etc. as possible is restored or enhanced.

165.19 Protection of Livestock and Wildlife

165.19.01 Protection of Livestock and Wildlife - Harassment of livestock or wildlife in and adjacent to the project site is prohibited. The presence of livestock or wildlife in or adjacent to the project site, field office trailers or construction camp shall not be encouraged by feeding. The Contractor is required to inform work crews of the location of wildlife and livestock crossing sites situated within the boundaries of the construction area.

If it is necessary to arrange the removal or transfer of beavers, bears or other wildlife from the work-site or camp the Contractor shall contact a British Columbia Ministry of Environment Fish and Wildlife representative prior to taking action, and follow the course of action recommended.

165.19.02 Disturbance of Fish and Wildlife - Rules regarding hunting, fishing and the discharge of firearms by the Contractor, employees and sub-contractors, within the

project area during the period of construction, shall be made in consultation with the Ministry of Environment Fish and Wildlife office. The Contractor shall be responsible for ensuring compliance with these rules.

165.20 Archaeological and Paleontological Discoveries - In the event that any item of particular archaeological, heritage, historical, cultural or scientific interest is found on the project site, such item(s) shall remain the property of the Province and the Contractor shall, on making or being advised of such a find, immediately cease operations in the affected area, minimize activities which create ground disturbance in and adjacent to the affected area, and notify the Ministry Representative and the Archaeology Branch of the British Columbia Ministry of Sustainable Resource Management. Work shall not resume within 30 m of the discovery site until an appropriate directive has been received from that agency.

To protect archaeological and paleontological sites that are situated within or adjacent to a project site, the Contractor may be required to use a variety of mitigative measures, including but not limited to drainage or erosion control, slope stabilization measures, or erecting fences or other suitable barriers to protect archaeological or paleontological sites that are situated within or adjacent to a project site. The costs associated with such mitigative measures will be borne by the Ministry, unless otherwise specified in the Contract.

A buffer zone, in which no land alteration or other activity is permitted, may be required to ensure adequate site protection. The width of this buffer zone shall be determined by the Archaeological Branch of the British Columbia Ministry of Sustainable Resource Management in consultation with a representative of the local First Nation(s) The Contractor shall be responsible for the actions of employees and sub-contractors with respect to site vandalism and the unlicensed collection of artifacts from designated archaeological sites in and around the construction area.

165.21 Resolution of Disputes - In the event that a dispute arises between the Contractor and the Ministry, or the Environmental Monitor and the Ministry regarding environmental matters related to the project, the Ministry, or where appropriate, the Environmental Agencies, will have the final decision.

SECTION 194

TRAFFIC MANAGEMENT FOR WORK ZONES

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

PART A – GENERAL

194.01 General - Traffic management requirements will be as specified herein and as specified in the Contract.

The Contractor's Traffic Management Plan, Construction Schedule and its provisions for traffic control and safety will be based on providing safe passage through the Work Zone for all road users (i.e. cyclists, pedestrians and motorists). The Contractor will at all times make provisions for traffic through the Site to a sufficiently high standard to ensure the safety of the workers on the Project, the safety and the convenience of the travelling public, driver expectancy, and the protection of the Work Area. The Contractor will supply and maintain all traffic control devices and personnel necessary to provide all traffic control.

The Contractor's accommodation for pedestrian and vehicular traffic, and proposed traffic control procedures, for each work operation, will require approval by the Designer, Quality Manager and Ministry Representative prior to the start of that operation.

Any one or more of the advance warning area, transition area, buffer zone and termination area of the traffic control (work) zone may be outside the Limits of Construction, but this will in no way diminish the Contractor's responsibility to meet the requirements of the Traffic Management Plan.

194.02 Definitions - For purposes of this Section, the following general terms are defined in the Traffic Management Guidelines for Work on Roadways:

- Traffic Management Plan
- Traffic Management Strategy
- Traffic Control Plan
- Incident Management Plan
- Public Information Plan
- Implementation Plan

In addition, the following terms are used in this Section and defined below:

"Work Area" refers to an area that contains the work activity and is closed to traffic and set aside for the exclusive occupation by workers, equipment and construction materials. Work areas may remain fixed in location or may move as work progresses.

"Work Zone" as defined in the Traffic Control Manual for Work on Roadways.

"Construction speed zone" as defined in the Traffic Control Manual for Work on Roadways.

194.03 Standard Documents – Traffic Management Plans shall be developed in accordance with standards defined in the latest version of the following documents:

- MoT Traffic Management Guidelines for Work on Roadways (TMG)
- MoT Traffic Control Manual for Work on Roadways (TCM)
- MoT Electrical and Traffic Engineering Manual and all current Technical Bulletins
- TAC Geometric Design Guide of Canadian Roads
- MoT Supplement to TAC Geometric Guide
- Workers' Compensation Board (WCB) Regulation

All Ministry Publications listed above can be purchased from:

GOVERNMENT PUBLICATION SERVICES
MINISTRY OF MANAGEMENT SERVICES
PO BOX 9452 STN PROV GOVT
VICTORIA BC V8W 9V7

www.publications.gov.bc.ca/queries/service.html

Email: QPPublications@gems5.gov.bc.ca

PHONE: 1-800-663-6105
FAX: (250) 387-1120

194.04 Traffic Control Supervisor - The Contractor will designate a traffic control supervisor who has the Contractor's authority to respond to traffic control requirements and who will provide surveillance of traffic conditions through the Work Zone. This person or qualified designate will be available at all times, both when the Contractor is working and during weekends or other periods when the Contractor's forces are not active on the Project, to respond to calls from the police and from representatives of the Ministry concerning damage or deficiencies in the traffic control devices and the passage for traffic through the Work Zone.

The traffic control supervisor shall meet the requirements of the TMG and WCB Regulation and shall not be the Contractor's site superintendent.

194.05 Compliance with Acts and Regulations - The Contractor shall comply with the requirements of any legislation, Regulations and by-laws as are in force for the regulation of traffic or use of any road upon or over which it

is necessary to carry out work or to haul materials or things.

PART B – PLANNING AND SCHEDULING

194.11 Provide Traffic Management Plan - The Contractor shall prepare a Traffic Management Plan (TMP) in accordance with the Ministry definitions and guidelines specified in the Traffic Management Guidelines for Work on Roadways Manual.

The Contract will specify the Traffic Management Strategy and required documentation for the Traffic Management Plan.

The Traffic Management Plan will be fully integrated with the Contractor's plan and schedule for carrying out the work, comply with the requirements of the TCM, the Traffic Management Guidelines for Work on Roadways, the Traffic Management Strategies listed below, and with the provisions of the Contract.

194.12 Traffic Management Plan Review – The Ministry Representative may review the Contractor's Traffic Management Plan for general conformance with the requirements of the Contract for sufficiency of detail to provide a complete and unambiguous traffic control layout. No review or change requested by the Ministry Representative in any way relieves the Contractor of any if its responsibilities for ensuring safe and adequate traffic management. No work that effects traffic will be permitted until the Ministry Representative accepts the Traffic Management Plan.

Where the Traffic Control Manual shows a choice of traffic control layouts for a given specific situation, the Ministry Representative may request the Contractor to use an alternative layout which is acceptable.

The following time periods are required for plan review and acceptance unless otherwise specified in the Contract:

- A Traffic Control Plan shall be submitted to the Ministry Representative for review at least fifteen (15) working days prior to the start of any Works effecting traffic flows. The following conditions shall apply:
  - Any plan modifications shall be submitted to the Ministry Representative for review at least ten (10) working days prior to any changes being made in the field.
  - A staged plan shall be submitted to the Ministry Representative for review at least ten (10) working days prior to implementation of a new stage.
- The Implementation Plan, if required, shall be submitted to the Ministry Representative for review at least fifteen (15) working days prior to the start of any Works effecting traffic flows.
- The Public Information Plan, if required, shall be submitted to the Ministry Representative for review at least fifteen (15) working days prior to the start of any Works effecting traffic flows.
- The Incident Management Plan, if required, shall be submitted to the Ministry Representative for review at least fifteen (15) working days prior to the start of any Works effecting traffic flows.

PART C – TRAFFIC MANGEMENT OPERATIONAL DETAILS

194.21 Speed Zones - . Reduction in speed limits may take several forms as defined in the TCM 1.4.6. Construction speed zones within the Project will be determined at the discretion of the Road Authority or as specified in the Contract.

The Contractor's work plan, Construction Schedule and its provisions for traffic control and safety will include references to the reduced posted speeds referred to in DBSS 194.21.

The location and length of posted speed reduction zones shall be adjusted daily, or more frequently as appropriate to the Work, to provide the shortest reduced speed zone practical. For example, on a resurfacing project, the reduced speed limit would usually be invoked around active operations such as milling, priming, paving, and, if necessary, at access points to pits and the plant, but will not generally be acceptable for areas of the Site used only for haul.

194.22 Use of Channelling Devices and Taper Rates - Traffic cones, temporary highway delineators, standard barricades or other marker devices such as wooden or metal saw horses, temporary marking tape, etc., shall be used in the following circumstances for channelling traffic:

a) from one lane to another or for merging traffic where a reduction in the number of lanes must be effected

b) around a hazard on the travelled way

c) through an area of unfinished construction where the intended traffic lanes are not self-evident

d) at any other location where, in the Designer's or Ministry Representative's opinion, channelling will ensure the safe passage of public traffic.

Minimum taper rates and spacing between cones or delineators shall be as shown in TCM.

194.23 Equipment on the Travelled Way - When working on the travelled way equipment shall be placed in such a position as to present a minimum of interference and hazard to the travelling public. Equipment units shall be kept as close together as working conditions will permit and preferably on the same side of the travelled way. Equipment shall not be left overnight on the travelled way but shall be

located so as to avoid the possibility of an accident.

194.24 Roadway Configuration - The Contractor shall arrange to keep interconnected lanes of the highway open to traffic at all times with the least amount of inconvenience to the travelling public unless otherwise authorized by the Ministry Representative.

194.25 Roadway Condition - The Contractor shall keep the travelled way well-graded, free of potholes and of sufficient width that the required number of lanes of traffic may pass. The subgrade shall be completed progressively in order that gravel surfacing may follow in an orderly manner. Prior to tearing up a portion of the existing travelled way the Contractor shall obtain Ministry Representative approval and shall also have available a sufficient supply of crushed gravel, or another approved product, to ensure a smooth riding surface during the work.

194.26 Closures – Road closures shall not be permitted until the Contractor requests and receives written permission from the appropriate authority in the Ministry of Transportation. This request shall be submitted in writing not less than fifteen (15) days in advance of the closure.

Upon receiving written permission for the closure, the Contractor must notify the local police, fire, ambulance, municipality, school board, and public transit at least seven (7) days in advance of the closure.

The public must be notified by local radio and newspaper at least (5) days in advance of the closure. The Contractor shall arrange any public announcement relating to the Contract and incorporating the Ministry logo through the Ministry Representative and Corporate Writing Services. Such advertisements will be made at the Ministry's expense. Notices or other advertisements of the Contractor unrelated to the Contract or where the Ministry's logo or name is not incorporated may be made by the Contractor at the Contractor's expense.

All costs related to the above shall be borne by the Contractor.

194.27 Detours —Where detours are required for the execution of the work, the Contractor shall provide well graded, gravelled detours or temporary roads to facilitate the passage of traffic around the restricted construction area. The Contractor shall be responsible for signs, maintenance of same and maintenance of the detour or temporary road. The Contractor shall consider the condition of the pavement used for detours and its impact on the safety and function of the detour. Milled surfaces upon which traffic is to run shall be clean and allow adequate drainage.

The minimum design requirements for any detours will be specified in the Contract.

Where the anticipated detour extends beyond the existing road surface, the Contractor will submit an engineering design to the Ministry Representative for review, a minimum of fourteen (14) days in advance of building the detour. The engineering design will include the design speed, the horizontal and vertical geometry, plus evidence that the design vehicle is accommodated for all movements and that all the requirements of this Contract are satisfied.

Where the anticipated deviation is minor, the Contractor may apply to the Ministry Representative, in writing, for approval to proceed without a formal design. Such approval does not relieve the Contractor of any of its responsibilities for provision of a safe site.

194.28 Access to Property - During operations the Contractor shall provide and maintain reasonable road access and egress to property fronting along or in the vicinity of the work unless other reasonable means of road access exists. The Ministry Representative will be the sole judge of what may be deemed reasonable road access.

194.29 Winter Lay-Up – In the event that the Work is stopped due to winter conditions, the Site is to be left in a condition as specified in the Contract or as deemed acceptable by the Ministry Representative.

194.30 Delays – All major and minor delays to the travelling public will be tracked by the Contractor, documenting location, duration, queue length and delay incurred.

The Contractor will monitor the queue lengths created by any delay to the flow of traffic. Traffic must be cleared prior to starting another delay unless authorized by the Contract. If the traffic delays exceed those allowable under the Contract, the Contractor shall immediately cease construction activities and make all the travel lanes available to traffic as quickly as possible. Resumption of roadway operations shall be permitted as traffic levels dictate.

PART D – TRAFFIC CONTROL DEVICES

194.41 Traffic Signals - Short one-way sections, such as bridge and culvert construction detours, etc., may be controlled by traffic signals and it is acceptable where the restriction will be in effect for an extended period. The use of signals on long one-way zones is not recommended as delays to traffic may be of considerable length and might not be accepted by the motorist. No signal installation shall be permitted without the approval of the Ministry Representative.

Where traffic signals are requested by the Contractor, the Contractor shall arrange to have a qualified Traffic Engineering consultant prepare a Traffic Engineering study for the signal. The study shall document the following:

- Site layout
- Anticipated volumes
- Requirements for advance warning signs
- Actuation
- Proposed traffic signal timing plans

The Contractor will submit the study to the Ministry Representative for review. The approval for the installation of the signals will be determined by the Ministry Representative.

If the proposed traffic signal installation is approved by the Ministry Representative, the Contractor shall:

- Prepare a traffic signal design, signed and sealed by an electrical engineer registered with the APEGBC
- Supply and install all traffic controller equipment
- Supply power
- Operate and maintain the signals

Changes to the signal timing plans shall only be prepared by a qualified professional traffic engineer experienced in traffic signal timings. Traffic controller technology used shall comply with industry standard practice and shall have a conflict monitoring mechanism that will ensure signal malfunctions/conflicts result in the signal going to a flashing red state. Signals shall be vehicle actuated unless otherwise approved by the Ministry Representative. Signal operations shall conform to TCM 2.2.8 and the *Electrical & Traffic Engineering Manual*.

Where the work activity impacts the operation of an existing signal, the Contractor must submit a written notification to the Ministry Representative a minimum of ten (10) days in advance of the disruption to the signal.

194.42 Pilot Cars – The use of pilot cars may be warranted where public traffic must use a particularly hazardous route, or where traffic is required to remain in one lane, or change periodically from one lane to another, or negotiate a section of construction at restricted speed. Pilot cars shall be operated in conjunction with flag persons stationed at each end of the restricted section. When required, the Contractor shall provide sufficient pilot cars, as may be required from time to time, to keep a steady and controlled flow of traffic moving around or through the construction area.

Pilot cars shall be equipped with the following:

a) A sign or signs with minimum 200 mm letters, designating the vehicle as a pilot car, and showing the message in both directions. Signs shall be illuminated during the hours when headlights must be used. Signs shall not be displayed when the vehicle is not being used for piloting.

b) Amber flashing lights mounted on the roof at either side so as to be clearly visible from both directions. The flashing lights shall be in use for both day and night piloting.

194.43 Use of Flag persons – It is generally expected that flag persons will be required in the following situations:

a) when public traffic is required to pass working vehicles or equipment which may block all or part of the travelled roadway.

b) when it is necessary to institute a one-way traffic system through a construction area or other blockage where traffic volumes are heavy, approach speeds are high, and a traffic signal system is not in use.

c) in high speed, high volume areas where temporary protection is required while other traffic control devices (barricades, cones, signs, etc.) are being erected or taken down.

d) for emergency protection when other traffic control devices are not readily available.

e) in all situations, where adequate protection for workers, working equipment and public traffic is not provided by other traffic control devices.

194.44 Traffic Signs – All standard signs, new and replacement, shall meet current Ministry specifications. The Ministry *Specifications for Standard Highway Sign Materials, Fabrication and Supply* are located on our Ministry web page at::

www.th.gov.bc.ca/publications/eng_publications/geomet/geometsigns.htm

Prior to the commencement of work the Contractor shall supply all posts, hardware and equipment required for the proper installation and maintenance of the project (C-35) signs and "Construction Ahead" (C-18) signs. The Contractor shall place the signs as indicated in the TCM or as directed by the Ministry Representative.

Upon completion of the project the Contractor shall remove the C-35 signs and the C-18 signs, posts and hardware when directed by the Ministry Representative. The removal of the signs shall also include the backfilling, compacting and dressing of the postholes to the satisfaction of the Ministry Representative.

The signs shall be erected in advance of all highway and bridge construction projects whether undertaken by contract or day labour. The signs may be omitted at the direction of

the Ministry Representative if the project is not in public view.

The Contractor shall supply and maintain all necessary signs and traffic control devices required by the Design, this specification and the TCM. All signs and devices used must conform to the standards detailed in this manual.

194.45 Barricades, Lights, Delineators, Traffic Lines - The Contractor shall provide, erect and maintain all types of temporary marker devices including traffic line marking tape, barricades, fences, delineators, flare-pots, battery operated amber flashing lights and 110 volt steady burning or simultaneous flashing lights, in order to ensure safety to the workers and the general public. These devices shall be as required by the Contract and the TCM.

Flashing amber lights, if deemed necessary by the Designer, are used for extra night time effectiveness and the batteries of the flashing lights shall be checked each day.

Battery operated flashing lights, if two-faced, and if used to mark the right hand edge of a travelled lane, must have one face covered so that the light is not visible to traffic travelling in the other direction. Flashing lights having two faces showing in opposite directions may be used only if it is necessary that the obstruction or lane edge being marked be seen from both directions of travel. They should not be used in a line of delineators through an area of incomplete construction except where traffic must be diverted and a specific hazard exists.

Temporary road construction lighting installations shall be positioned as shown in FIGURE 1: Positioning of Temporary Lighting Adjacent to a Highway to minimize glare and resulting impaired driver vision. Lights shall be aimed away from any approaching traffic and at an angle of less than 45 degrees above vertical. Where lights cannot be located as shown then temporary glare screens shall be used to shield the traffic from the light source. These glare screens shall be metal or an opaque plastic material designed to shield the lighting from the eyes of the driver.

The Contractor shall be responsible for the application and removal of all temporary pavement marking and reflective devices. When traffic lanes have to be redefined for Long Duration Work, the Contractor shall eradicate all redundant temporary or pavement markings that are not required for the intended traffic patterns.

Class I and Class II standard barricades are illustrated in the TCM. Temporary delineator posts shall be as shown on SS Drawing SP194.02. In speed zones of 50 km/h or less, the minimum height of traffic cones shall be 450 mm. Where the maximum speed limit is over 50 km/h, the minimum height of cones shall be 700 mm.

194.46 Changeable Message Signs –The use of changeable message signs (CMS) may be specified in the Contract. Each sign must be portable. When in operation, the CMS shall be a minimum of two (2) metres from the bottom of the sign display to the road surface, and shall be level and capable of pivoting for sighting purposes. The specifications of the sign are as follows:

Sign unit:	Yellow/Orange LED display
Sign display:	3 lines with 8 characters per line
Character size:	450mm (18 inches)
Character matrix:	5 x 7
Remote dial-up access:	By cellular phone or equivalent

A full matrix sign may be used given that it has the display parameters noted above.

The Contractor is responsible for the CMS including but not limited to bringing the signs to the Project, moving and setting-up, providing storage as required including moving the signs into and out of storage, maintaining the signs in full operating condition, ensuring the correct message is being displayed, damage to or loss of the signs and returning the signs to the supplier. The Contractor shall provide to the Ministry Representative with a description of the messages displayed and the procedures of how the messages on the CMS will be changed when the Contractor is on and off site.

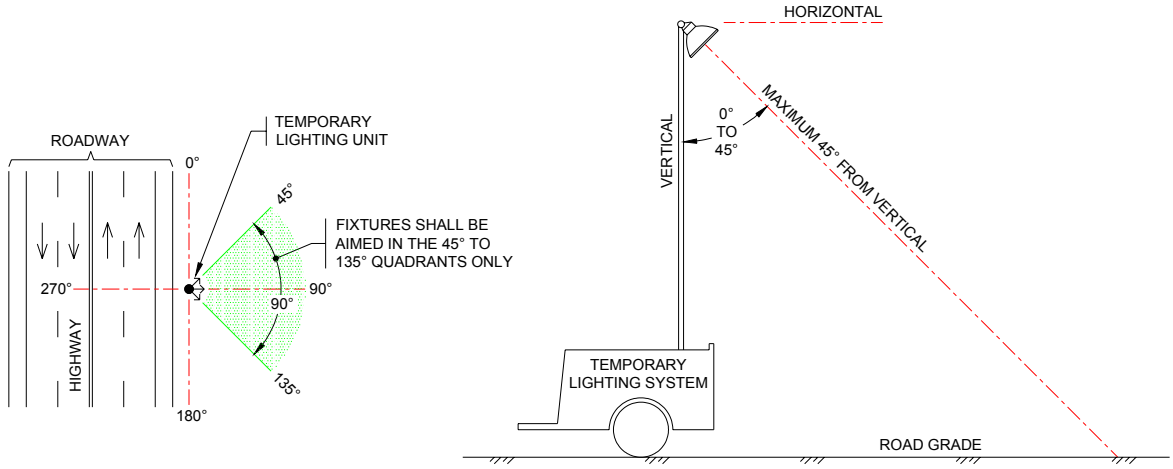


FIGURE 1: Positioning of Temporary Lighting Adjacent to a Highway

PART E – GENERAL MAINTENANCE

194.51 Maintenance - The Contractor shall be responsible for the continuing maintenance of all traffic control devices in use as follows:

a) Signs, etc., not applying to existing conditions shall be removed or covered. If covered, such as after sundown or during periods when work is suspended, the covering shall be placed over the sign so as to totally obliterate the message thereon. Where operations are carried out in stages, only those devices or signs, which apply to the current stage, are to be left in place or uncovered.

b) Signs shall be checked daily for legibility, damage, suitability and location. Dirty, damaged, unsuitable or misplaced signs shall be cleaned, repaired or replaced. Signs and delineators with a reflective surface shall be cleaned as frequently as necessary to ensure full reflectivity. As considerable damage may be inflicted on signs and devices during a weekend, the Contractor shall conduct a check for replacement or repairs before the rush hour on Monday morning. Similarly, high traffic volume and recreational routes shall be checked during the weekend or at any other time that works is not in progress.

During hours when headlights are required, signs that have not been treated with a reflective coating shall be illuminated. After sundown all signs shall be checked for visibility and those that cannot be seen clearly shall be adjusted or replaced.

SECTION 200

CLEARING AND GRUBBING

200.01 Clearing - Except for trees and shrubs to be preserved as indicated on the Drawings or designated by the Designer, "Clearing" shall mean the complete removal and disposal of all standing and fallen trees, stumps, logs, upturned roots, rotten wood and all other vegetation growth, and accumulations of rubbish of whatsoever nature, and any other objectionable material from the area shown on the Drawings, or as directed by the Designer. Protective measures for the protection and retention of vegetation are specified in DBSS 769.

It shall also include the removal of existing cribwork, guard rail, fences, siphon structures built of timber and/or metal, culvert pipes of all types and timber box culverts within the limits of the work; plus removal of concrete buildings or their foundations, concrete bridge piers or abutments and their footings, concrete siphon boxes or concrete box culverts.

The area to be cleared shall include those areas required for the highway right-of-way, borrow pits, quarry sites, gravel pits and access roads thereto, if ordered by the Designer; off-take ditches or other drainage ditches and channels for stream diversions either within or without the right-of-way; clearing of debris in streams as indicated on the Drawings, or as directed by the Designer within the Site; foundations for structures and the right-of-way as shown on the Drawings, or as directed by the Designer, for intersecting public and private road approaches or diversions of same outside the main highway right-of-way. All such areas are hereinafter referred to as the "Area."

200.02 Disposal of Material - All material, slash and debris resulting from clearing operations shall be disposed of by burning unless there are specific provisions or open burning regulations for otherwise disposing of same. Unmerchantable timber, stumps, etc., shall not be disposed of by pushing outside the clearing and grubbing right-of-way. All trees shall be felled within the Area but, in the event of any trees falling outside the Area, such trees shall be cut up and together with all debris and slash therefrom, disposed of off-Site or, where permitted by the Contract and burning regulations, brought back to the Area and there burned. The Contract or the Designer may designate certain trees or shrubbery to be left standing, in which case the Contractor shall take every precaution not to damage or injure such trees or shrubbery in felling adjacent timber, burning or any other clearing operations as specified in DBSS 769.

Such trees or shrubbery are to be limbed or thinned to such height and extent as may be directed by the Designer.

Except as hereinafter provided, all slash and debris shall be disposed of off-Site or, where permitted by the Contract and burning regulations, piled and burned at points located centrally in the Area. The number of fires to be started at any one time shall be limited to the capacity of the Contractor's equipment and organization to provide adequate protection against the spreading of the fires to adjacent timber or property.

All burning shall be carried out subject to the provisions of the Open Burning Regulation, the Forest Act and regulations thereto. In the event of regulations prohibiting burning, the slash, debris, etc., will be disposed of off-Site or may be piled along the sides of the Area and burned at a later time when permitted by the regulation. When the regulations do not allow burning prior to the Completion Date, the material shall be disposed of off-Site prior to the Completion Date.

200.03 Grubbing - Grubbing means the entire removal and disposal by burning as described in DBSS 200.01 and DBSS 200.02 of all stumps, roots and embedded logs to a depth of 0.6 m below the ground line or other such deeper depth as required by the Designer.

Grubbing shall be carried out over the entire Area as defined in DBSS 200.01. Grubbing shall be fully completed in advance of grading operations.

200.04 Danger Trees - This item shall consist of removing any leaning or dangerous trees or snag outside the clearing Area.

Such trees shall be felled and disposed of as provided for in DBSS 200.01 and DBSS 200.02,

The Contractor will not be required to grub the stumps of any danger trees unless required by the Designer.

200.05 Not Used

200.06 Mail Boxes - Where it becomes necessary to remove or otherwise disturb existing mail boxes within the limits of any project, the Contractor will be required to install the boxes temporarily in such a position that their usefulness will not be impaired. After construction work has been completed, the boxes shall be re-installed at positions designated by the Designer and acceptable to Canada Post. No new posts or materials will be required, except that any damage caused by the Contractor to either boxes or supports, due to the Contractor's negligence, shall be repaired by the Contractor at the Contractor's expense.

200.07 Removal of Existing Signs and Symbols When construction operations are started, the Contractor shall remove all existing Signs and Symbols of the British Columbia Ministry of Transportation not needed for the safe and orderly direction of traffic, and any other authorized signs within the limits of the right-of-way, and shall deliver them to such point as directed by the Ministry Representative.

SECTION 201

ROADWAY AND DRAINAGE EXCAVATION

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

201.01 General - Roadway and Drainage Excavation shall include all excavation and the construction of all embankments required for the formation of the roadbed, parking areas and look-outs; excavation for any drainage ditch, off-take ditch or channel for stream diversion within or outside the right-of-way; removal of surcharge material, topsoil and organic waste material from the roadway, and of topsoil and unsuitable overburden from any borrow pit or any available Ministry or Crown gravel pit, and disposal of same, as may be directed by the Designer; excavation of materials below grade; excavation of borrow pits; the grading necessary to construct any frontage road, gravel pit access road, borrow pit access road or any other access road, public or private,; the grading of connections to intersecting roads, public or private,; the hauling and disposal of all excavated material, and the trimming and shaping of all excavations and embankments. All Works will be confined to the right-of-way except where agreements for access rights are in place for other lands, public or private.

201.02 Extent of Work - The dimensions of the excavations and embankments shall be as shown on the Drawings or the Design but the Designer may increase or decrease the width of the roadbed or the ratio of the slopes as conditions and circumstances may necessitate.

Construction Grade is defined as the grade upon which the Asphaltic Pavement is laid.

Subgrade is defined as the grade upon which the first layer of select granular sub-base or base material is laid.

MATERIALS

201.11 Description of Material Types - Excavation shall be classified under the following material types:

a) Type A
b) Type B
c) Type C
d) Type D

The material encountered in any highway excavation shall be classified as one of the four types listed above.

In no case shall a material be classified using percentages of two or more types.

It is possible that material may change at some level in the excavation. Where such a change occurs, sufficient field measurements will be taken to establish the boundary between the two types of material.

201.11.01 Type A - Solid Rock - Type A shall include all forms of "solid rock in place" occurring in masses, ledges, seams or layers of sufficient hardness to require breaking by continuous drilling and blasting before excavation and removal.

It shall also include detached masses of rock or boulders individually containing a volume of 1.5 m^3 or more.

201.11.02 Type B - Materials Containing More Than 50% Loose or Broken Rock - Type B Material shall mean loose or broken rock, boulder gravel, or any other material containing at least 50% boulders or broken rock, having a size greater than 0.03 m^3 but less than 1.5 m^3 and in a state not requiring loosening, or drilling and blasting for removal.

Any excavation or portion thereof, containing the 50% boulders or broken rock of the size described above, shall be classified as this type of material and will be paid in its entirety under this category.

Any excavation, not containing 50% boulders or broken rock of the size described above, shall be classified in its entirety as Type D Material, and in that event no allowance will be made for the smaller percentage of boulders or broken rock in the material.

201.11.03 Type C - Materials Requiring Ripping - Type C shall include those materials of such density or so firmly cemented together that a force of at least 360 kN per linear metre of ripper shank or ripping equipment embedded in the material is required to loosen the material. If the material can be loosened with a lesser force, then the material will be classified as Type D.

If a means other than ripping equipment is used to loosen materials, the force generated must be equivalent to that specified above.

Where deemed necessary by the Quality Manager, a ripping test will be conducted using a bulldozer or other appropriate equipment, under suitable weather conditions, that will develop sufficient traction and effectively deliver a force of 360 kN per lineal metre of ripper shank embedded into the alleged Type C. The material will only be classified as Type C if it cannot be loosened under such an applied load.

201.11.04 Type D - All Other Materials - Type D shall include all other materials of a nature not included in the foregoing definitions of Type A, Type B, or Type C; regardless of the nature or condition of the material or the method used to excavate or remove.

Type D shall also include any unsuitable materials of an organic nature, unless such materials have been specifically set apart in the Design or the Special Provisions of the contract.

201.12 Unsuitable Material - Unless otherwise shown, specified or directed, all topsoil and organic material shall be stripped from the highway cut and fill areas.

Where additional material is found above or below subgrade in an excavation or in the foundation of an embankment, which in the opinion of the Designer or Quality Manager is unsuitable, such material shall be removed. Soft clays and peats are examples of normally unsuitable materials.

Backfilling, where required, shall be carried out with approved materials, as ordered by the Designer or Quality Manager..

Material that is unsuitable for the construction of embankments shall be disposed of in a geotechnically and environmentally safe manner.

Upon completion of the road grade, topsoil and organic material suitable for growth of ground cover shall be spread on the cleared right-of-way and up the fill slopes to generally blend in with the cross-section, to a nominal depth of 200 mm. Care shall be taken that proper drainage is maintained. This topsoil shall be trimmed to a neat appearance preparatory to seeding by others. Surplus organic materials shall be disposed of off-Site or on-Site with the approval of the Ministry Representative

CONSTRUCTION

201.31 Rock Cuts- See DBSS 204

201.32 Dangerous, Loose and Overhanging Rock - The Contractor shall be required to remove dangerous, loose and overhanging rock within or outside the limits of the highway right-of-way, to ensure the safety of the Site and the public

201.33 Not Used

201.34 Not Used

201.35 Not Used

201.36 Rock Embankments - This section applies to embankments constructed from material containing more than 15% by volume of rock larger than 150 mm.

Embankments shall be constructed in layers equal in thickness to the largest size of the material but not exceeding 0.7 m. Greater lift thicknesses will be permitted by the Designer under special conditions provided the Contractor can spread the larger material satisfactorily and achieve the design density and stability in the embankment.

The material shall be deposited and spread so that the larger rocks are well distributed and the intervening spaces are filled with smaller sizes as may be available to form a stable embankment. The finer portion of excavated rock shall be retained for the top transition layer. Each layer shall be compacted by routing the loading construction equipment over the entire width, supplemented by additional compaction equipment, as necessary to ensure a stable embankment.

When a rock embankment is overlain by an earth embankment or by sub-base or base course materials, the top 0.5 m of the rock fill shall be sealed with smaller rock particles and suitable soil materials, if necessary, to prevent the uppermost layers penetrating into voids in this rock embankment.

Where permitted by the Designer, and where the width is too narrow to accommodate equipment, material for the toe of the side hill rock fills may be placed by end dumping until sufficient width of the embankment has been formed to carry equipment, after which the remainder shall be placed in layers and compacted as specified. Any loose material left on the slopes by end dumping shall be incorporated in the fill and compacted.

When embankments are made on hillsides or where a new fill is to be applied upon an existing embankment, the slopes of the embankment or original ground (except rock) shall be terraced in a continuous series of steps a minimum of 1.5 m wide as the embankment rises.

The material from step excavation shall be spread and compacted into the adjoining embankment. No additional payment will be made for excavation or for placing step material in the adjoining fill.

201.37 Earth Embankments - This section applies to embankments constructed from material containing less than 15% by volume of rock larger than 150 mm. Embankments shall be constructed in such a manner that they shall be completely stable with reference to designated traffic loading.

The natural surface shall be excavated to remove organic soils or other unsuitable material, as directed by the Designer.

The earth embankment shall then be constructed in successive horizontal layers not exceeding 200 mm in loose thickness except that the top 500 mm shall be constructed in layers not exceeding 100 mm in loose thickness. Each layer shall be compacted to minimum 95% of the laboratory density obtained by the current ASTM test method D 698, except in the top 300 mm of the embankment, which shall be compacted to minimum 100% of the laboratory density.

The above maximum lift thickness requirements may be waived where the Contractor can:

- demonstrate that the proposed compaction equipment can uniformly compact a thicker layer;
- attain the required degree of compaction at all levels within the lift; and
- supply compaction testing equipment and services for Quality Control and Quality Assurance that will reliably test to the full depth of the proposed lift thickness.

No organic soils shall be placed in the embankment. Soils with high moisture content that cannot be compacted to the required density shall not be employed without prior aeration and drying.

When embankments are made on hillsides or where a new fill is to be applied upon an existing embankment, the slopes of the embankment or original ground (except rock) shall be terraced in a continuous series of steps a minimum of 1.5 m wide as the embankment rises.

Where suitable for embankment, the material from step excavation shall be spread and compacted into the adjoining embankment.

The Contractor shall be responsible for selecting equipment and methods of attaining the specified degree of compaction.

201.38 Frozen Material - The use of frozen material in embankments will not be permitted; the only exception is broken rock containing less than 15% passing a 4.75 mm sieve. Frozen excavated material which will be suitable when dry may be stored and allowed to thaw and dry, and then placed in the embankment.

201.39 Snow Removal - Snow overlying the surface of a cut, or the site or surface of an embankment shall be removed and deposited beyond the slope stakes.

201.40 Bridge End Fill – Material for bridge end fill shall be in accordance with DBSS 202.04 and DBSS 202.05. Construction of bridge end fill shall be in accordance with DBSS 202.23.

201.41 Transition Sections - Subgrade, other than solid rock, shall be subcut 1 m deep at the line of transition from cut to embankment. The subcut shall taper to zero depth 8 m within the cut. The embankment construction shall be carried back over the subcut only after the embankment has reached the level of the bottom of the subcut

201.42 Slopes and Slides - The slopes of all excavations and embankments shall be trimmed and maintained neatly and evenly to the line and slope indicated on the Design or as directed by the Designer.

No undercutting of slopes in will be permitted.

201.43 Ditches - Ditches which may be considered necessary for the proper drainage of the work shall be constructed at such points and to such cross-section, alignment and grade as the Designer may direct. This shall include inlets and outlets to culverts and ditching of all kinds. Ditching quantities will be considered as ordinary excavation quantities and will be measured, classified and paid for as such.

201.44 Borrow - Borrow shall consist of the excavation and disposal of suitable and satisfactory material obtained from borrow pits for the construction of embankments, subgrade or shoulders.

Borrow pits shall be so excavated that they will drain to the nearest natural outlet, or to an outlet designated by the Designer. Side slopes of borrow pits in all cases shall be dressed as required by the Designer to obtain stable slopes..

201.45 Not Used

201.46 Watering - Water for compacting embankments, constructing subgrades and surfacing, and for laying dust caused by grading operations and traffic, shall be applied in the amounts and at the places designated by the Quality Manager. Water shall be applied by sprinkling with tank trucks equipped with spray bars and suitable control apparatus

The Contractor shall make all necessary arrangements for obtaining water.

201.47 Finishing of Roadway - Before acceptance and final payment is made, the entire roadway, including the roadbed, shoulders and ditches shall be neatly finished and trimmed to the lines, grades and cross-sections shown on the Drawings, or as directed by the Designer, to reproduce smooth surfaces, slopes and a uniform cross-section. Subgrade shall be finished within a tolerance of ± 15 mm except for rock cut fills where the tolerance shall be ± 50 mm.

All drainage ditches, waterways and culverts shall be opened up and cleaned out to restore them to their full effectiveness.

All loose rock and boulders within the right-of-way resulting from grading and grubbing operations shall be gathered up and buried, or otherwise disposed of.

The Contractor shall grade all portions of the right-of-way outside the Excavation and Embankment areas to conform to the general ground line.

201.48 Surcharge Pre-consolidation -
Pre-consolidation by surcharging shall be carried out where indicated on drawings and/or Special Provisions, or as directed by the Designer. Embankments on these areas may be placed directly on the natural ground without removal of the organic materials, only if so directed by the Designer

The Contractor may place the initial lift of embankment to a minimum depth required to carry the Contractor's hauling equipment. The remainder of the embankments shall be constructed in accordance with procedures set forth in this Section; except that compaction will not be required on the surcharge material above construction grade.

To avoid shear failures, the rate of construction shall be rigidly controlled by instrumentation installed by the Contractor.

The surcharge shall remain on the embankments for a period determined by the Designer, and adjusted as may be indicated by the instrumentation.

Should any of the instrumentation become damaged or rendered unusable by the construction operations, same will be replaced by the Contractor.

201.49 Back Slope Stabilization - Drainholes, rock bolts, mesh and/or shotcrete may be required where rock is being excavated.

The backslope stabilization requirements will be assessed by the Designer during construction and the amount and location of the drainholes, rockbolts, mesh and shotcrete may be changed or deleted according to the condition of the rock encountered in the field.

SECTION 202

GRANULAR SURFACING, BASE AND SUB-BASES

PART A - SCOPE OF SPECIFICATION

202.01 Preliminary and General - This specification describes the classification, the materials and the professional standards required for the production of granular aggregates and the construction of granular surfacing, granular base and select granular sub-base.

PART B - MATERIALS

202.02 Aggregate Classification

202.02.01 Crushed Surfacing Aggregates:

i) High Fines Granular Surfacing Aggregate (HFSA): HFSA is used for surfacing of gravel roads and when paving is not scheduled for a few years. Cohesion for this aggregate is achieved by plastic fines.

202.02.02 Crushed Base Course Aggregates: These are high strength, high load bearing, high quality, crushed granular base materials. As these materially are not well suited for long-term traffic, they are usually overlaid by surface course materials (HFSA or asphalt/concrete pavement).

i) Well Graded Base (WGB): WGB is the standard base course material used for pavement structures, well graded with moderate permeability.

ii) Intermediate Graded Base (IGB): IGB is intermediate in gradation and permeability between well graded aggregate and open graded aggregate.

iii) Open Graded Base (OGB): OGB is a highly permeable, gap graded aggregate.

202.02.03 Sub-Base Aggregates – A granular material placed between the sub-grade and the base course aggregates, providing strength to the pavement structure.

i) Select Granular Sub-Base (SGSB): SGSB is the standard aggregate, crushed or screened, placed directly above subgrade, with moderate permeability and a wide gradation.

ii) Intermediate Graded Sub-Base (IGSB): IGSB is a higher permeability, crushed sub-base aggregate.

iii) Open Graded Sub-Base (OGSB): OGSB is a highly permeable, crushed sub-base aggregate.

202.02.03 Bridge End Fill (BEF): BEF is quality granular fill placed behind and below a bridge abutment to provide good drainage, a smooth transition from the bridge approaches to the bridge structure, and a suitable material through which to drive piles.

202.03 General Description of Work - The aggregates for granular surfacing, base and sub-base may be supplied either by the Contractor or produced from a Ministry provided source. Production of granular surfacing and granular base requires crushing and placing in stockpile. Production of select granular sub-base may be by crushing, screening or direct excavation.

Granular aggregate shall be loaded from the stockpile, crushing and screening plant, or pit face, as may be applicable; hauled out to the highway, placed accurately without segregation and thoroughly compacted to established line and grade.

The Contractor shall maintain the prepared surface of the base course aggregate until it has been paved or until completion of the contract, whichever is applicable.

202.04 Aggregate Quality

202.04.01 General - Granular aggregate shall be composed of inert, clean, tough, durable particles of crushed rock, gravel, sand and fines capable of withstanding the deleterious effects of exposure to water, freeze-thaw, handling, spreading and compacting. The aggregate particles shall be uniform in quality and free from clay lumps,

wood and free from an excess of flat or elongated pieces.

Aggregates for surfacing, base, sub-base and bridge end fill will be tested in accordance with the Table 202-A:

TABLE 202-A AGGREGATE QUALITY TESTS

BASED ON ASTM	TITLE OF TEST
C 136	Sieve Analysis of Fine and Coarse Aggregates
C 117	Wash Test of Aggregates
C 88	Soundness of Aggregate by Use of Magnesium Sulphate.
D 2419	Standard Test Method for Sand Equivalent Value of Soils and Fine Aggregate
D 6928	Abrasion Loss by Micro-Deval
D 4318	Plastic Limit Determination
C 295	Standard Guide for Petrographic Examination of Aggregates for Concrete
DBSS 202 Appendix 1	Fracture Count on Coarse Aggregate

202.04.02 Primary Quality Tests - All aggregates for surfacing, base, sub-base, and bridge end fill shall meet the requirements of Table 202-B.

TABLE 202-B AGGREGATE PROPERTIES

TEST	Test Result							
	Surfacing Aggregates	25 mm Base Course			Sub-Base Aggregates			Bridge End Fill
	HFSA	WGB	IGB	OGB	SGSB	IGSB	OGSB	BEF
Sand Equivalent	≥ 20	≥ 40	≥ 40	≥ 40	≥ 20	≥ 20	≥ 20	≥ 20
Micro-Deval loss factor	≤ 25%	≤ 25%	≤ 25%	≤ 17%	≤ 30%	≤ 25%	≤ 25%	≤ 30%
Fractured Faces Method "A"	≥ 50	≥ 50	≥ 50	≥ 50	n/a	≥ 50	≥ 50	n/a
Plasticity	≤ 6	n/a	n/a	n/a	n/a	n/a	n/a	n/a

Where the Ministry has pre-existing test results, the Ministry shall make this information available to the Contractor and the Contractor will be able to rely on the data and not have to perform the corresponding tests on materials extracted from the investigated portions of the pit or quarry.

In circumstances where historic performance of material from a source has proven satisfactory, the Ministry may,

waive any or all of the aggregate testing required in DBSS 202.04.

202.04.03 Secondary Quality Tests - All aggregates for surfacing, base, sub-base, and bridge end fill failing any primary quality test indicated in DBSS 202.04.02 shall further be tested as follows to determine acceptability or, at the Contractor's option, a new source shall be found.

i) If the Micro Deval test results show that the material fails, then the soundness shall be reassessed by use of Magnesium Sulphate (ASTM C 88). The material will be considered acceptable if the loss after 5 cycles is less than:

- 20% for coarse aggregate (the material retained on or above the 4.75 mm sieve)
- 25% for fine aggregate (the material passing the 4.75 mm sieve)

ii) If the aggregate fails any of the Sand Equivalent, Micro-Deval or Magnesium Sulphate tests, it shall be examined using the Standard Guide for Petrographic Examination of Aggregates for Concrete (ASTM C 295) to determine the cause of failure. The Contractor's Quality Manager shall review the test results and, with the approval of the Ministry Representative, determine whether the material is acceptable or not.

202.05 Aggregate Gradation

Gradation shall be determined in accordance with ASTM C 136 (Dry Sieve) and, for the material passing the 0.075 mm sieve, ASTM C 117 (Wash Test).

202.05.01 Aggregate shall have a gradation that defines a curve (% passing versus log sieve size) with a slope between adjacent sieves, equal or intermediate to the corresponding slopes of the boundary curves defined by the specification. Gradations shall fall within the limits, for the specified classification, shown in Table 202-C.

TABLE 202-C AGGREGATE GRADATIONS

SIEVE SIZE	Percent Passing (%) Sieve Size							
	Surfacing Aggregates	25 mm Base Course			Sub-Base Aggregates			Bridge End Fill
(mm)	HFSA	WGB	IGB	OGB	SGSB	IGSB	OGSB	BEF
75	---	---	---	---	100	100	100	100
50	---	---	---	---	---	55 – 100	70 – 100	30 – 100
37.5	---	---	---	---	---	40 – 80	50 – 85	---
25	100	100	100	100	---	---	---	---
19	85 – 100	80 – 100	65 – 100	75 – 100	15 – 100	17 – 40	15 – 55	20 – 100
9.5	60 – 85	50 – 85	---	30 – 65	0 – 100	---	---	---
6.3	---	---	---	---	---	---	0 – 20	---
4.75	40 – 70	35 - 70	30 - 70	---	---	---	---	10 – 60
2.36	---	25 - 50	15 - 40	5 - 30	---	10 – 25	0 – 10	---
1.18	20 – 50	15 - 35	10 - 30	0 – 10	---	---	---	6 – 32
0.600	---	---	---	---	0 - 100	---	---	---
0.300	10 – 30	5 – 20	5 – 15	0 – 8	0 – 15	4 – 15	0 – 8	4 – 15
0.075	5 – 15	0 – 5	0 – 5	0 – 5	0 – 5	0 – 5	0 – 5	0 – 5

202.05.02 Rut Resistance – Any aggregate supplied shall, in addition to meeting the gradation requirements specified above, not rut when proof rolled with a fully loaded tandem axle truck. Any aggregate which does rut shall be removed and replaced, or blended with suitable aggregates, to meet both the gradation requirement and the rut resistance requirement.

PART C - USE OF MINISTRY-PROVIDED PITS OR QUARRIES

202.06 General – Ministry-provided pits and quarries are those identified in the Contract which the Ministry is offering to the Contractor as potential aggregate sources.

They may include pits and quarries:

a) owned by the Ministry

b) leased by the Ministry, or

c) attained by the Ministry through special agreement with the landowner.

Use of such pits and quarries shall be at the Contractor's option unless the Contract clearly state that the source shall be used for specific purposes.

When a Ministry pit or quarry is provided for the Contractor's optional use under the Contract, the Contractor, by entering the pit or quarry, will be deemed to have accepted all obligations, risks and costs for the production of aggregate meeting the requirements of Part B of this section. The Ministry gives no warranty that its pit or quarry will meet the quality and quantity requirements.

Where a pit or quarry is designated as a mandatory source, the Ministry waives the aggregate quality requirements of DB SS 202.04.02 for aggregates required by the Ministry to be produced from that source.

202.07 Source of Supply - The Contract documents may identify one or more pits or quarries, for use as aggregate sources at the Contractor's discretion.

The Ministry will provide those gravel pits or quarries and the right-of-way to those pits or quarries and will pay the cost of royalties and fencing. The cost of constructing and maintaining access roads shall be considered as part of the cost of producing the construction aggregates and shall be payable by the Contractor. The Ministry will exercise its best judgement in the selection of the pit or quarry sites; however, the failure of the pit or quarry to yield materials of uniform quality shall not constitute grounds for a claim by the Contractor against the Ministry.

The Contractor shall be satisfied as to the gradation and other characteristics of the raw material in the Ministry provided pit or quarry and as to the nature and amount of work required to produce materials which shall meet all gradation requirements, and the Contractor shall, at the Contractor's expense, remove any objectionable materials from the aggregate.

202.08 Development of Pit or Quarry - Before any material for aggregate is excavated from the pit or quarry, the site shall be cleared and grubbed by the Contractor, and all debris disposed of to the satisfaction of the Ministry Representative. If topsoil or undesirable overburden exists, these materials shall be stripped and placed in separate stockpiles to the satisfaction of the Ministry Representative for future use in pit or quarry reclamation.

The Contractor shall take adequate precautions and actions to prevent pollution of the air, watercourses and adjoining land by the stockpiled material. The Contractor shall provide any protection necessary to protect the stockpile from erosion.

Watercourses located within or immediately adjacent to the gravel pit or quarry operations are to be protected against pollution. Where it is necessary to divert watercourses, application shall be made to Ministry of Environment, or the appropriate Federal agent, prior to such diversion being undertaken. Upon termination of operations, the watercourses shall be restored by the Contractor to as close to their original condition as is practicable.

The Contractor shall not operate the pit or quarry in a manner which will contaminate remaining granular material nor leave the pit or quarry in a condition which will limit its future use.

The Contractor shall follow the pit development plan, provided by the Ministry Representative. Stockpile sites in

Ministry-provided sources shall be per the pit development plan.

202.09 Use of All Materials - Any Ministry provided pit or quarry, once entered and developed, shall be used to its full potential. Normally, the Contractor shall produce select granular sub-base by selection or screening, provided that oversize reject aggregate is removed and properly placed in a stockpile for later crushing and use. However, if 5% or greater oversize occurs, pit run aggregate shall be crushed to produce 75 mm crushed granular base for use in place of select granular sub-base.

For the production of crushed surfacing and base aggregates, as well as IGSB and OGSB, the Contractor shall provide crushing equipment such that all aggregate which will pass through 375 mm x 450 mm slotted openings shall be used for the production of crushed aggregate; rocks which will not pass through these openings shall be stockpiled or disposed of to the satisfaction of the Ministry Representative. Crushing and screening equipment shall be provided with adequate facilities and capacity to bleed off reject aggregate in usable condition, or otherwise remove any excess of fine aggregate, dust or objectionable aggregate coatings that may be present in or on the aggregate, so as to make it generally acceptable for use. No portion of the products of crushers or screening plants that can be used shall be wasted. .

202.10 Unsuitable Materials - The rejection or addition of any particular size material necessary to produce gravel conforming to specification requirements shall be the responsibility of the Contractor.

202.11 Change of Ministry Provided Pit or Quarry - If the Ministry Representative requires a move to a new Ministry provided pit or quarry the Contractor shall l be given notice in writing The Contractor will not receive any allowance or compensation to cover such a move.

202.12 Pit Cleanup - When the Contractor discontinues operations in any pit or quarry, the Contractor shall trim the sides of pit excavations, waste piles and stockpiles to a 1.5H to 1V slope, or such other slope as the Ministry Representative may direct, open up such drains or ditches as may be required to prevent water standing therein, and leave the site in a neat condition, all to the satisfaction of the Ministry Representative. The trimming, sloping, ditching and draining of the pit or quarry, as well as maintenance of the access roads, shall be considered part of the cost of producing the granular aggregate.

If a pit has been fully depleted and is to be abandoned, the Contractor may be requested to reshape the sides to a slope not steeper than 2H to 1V using the stockpiled overburden if necessary, and cover same with any topsoil previously removed and stockpiled, all to the satisfaction of the Ministry Representative. In such cases, the replacing of stockpiled overburden and topsoil will be paid for in accordance with DBSS 202.32.

202.13 Work in Ministry Pits or Quarries – All work in a Ministry pit or quarry will be in accordance with DBSS 145.26.

PART D - CONTRACTOR SUPPLIED AGGREGATES

202.14 Preliminary and General - Aggregates, when supplied from private pits or quarries, shall meet the requirements of DBSS 202, Part B.

202.15 Development and Clean Up of Private Pits or Quarries - Pit or quarry development and cleanup shall be in accordance with local and Provincial Regulations.

202.16 Contractor to Pay All Costs – The Contractor shall bear and pay all costs including applicable royalties for provision of granular aggregates supplied from non-Ministry-provide sources. The Contractor shall also pay all costs for any move of the equipment that may be ordered or required because a pit or quarry of the Contractor's choosing proves unsuitable for production of the contracted quantity of granular aggregates meeting the

requirements of DBSS 202.

PART E - PRODUCTION OF CRUSHED AGGREGATE

202.17 Equipment and Plant - The Contractor shall provide and maintain in good operating condition, all plant and equipment necessary to comply with the requirements of this specification. Such plant and equipment shall be of a type and capacity to ensure proper construction and maintenance of access roads, development of the pit or quarry, maximum use of the pit or quarry, production of granular aggregates to the standard of uniformity required, and adherence to the construction schedule.

Adequate sampling facilities shall be installed by the Contractor at the discharge end of the production conveyor. The Contractor shall ensure that the plant and equipment meet the requirements of the current Pollution Control Act and Workers' Compensation Board Regulations.

202.18 Working of Pit - A pit shall be worked in such a manner as to ensure that granular material excavated and fed to the crushers, screening plants or loaders shall be as uniform as possible. This may require the use of equipment capable of deep excavating through horizontal gravel layers or carrying out selective digging or cross blending over a large area of the pit. Before commencing production, the Contractor's quality management plan shall describe the proposed method of production.

202.19 Crushing Operations - The Contractor shall adjust the crushers and/or screening plants and provide such screens as may be necessary to produce and maintain acceptable gradations of granular aggregates.

202.20 Crusher Gradation Control

202.20.01 Design Aggregate Gradation

For each aggregate gradation classification specified for production, field and/or laboratory tests shall be performed by the Contractor to select a single gradation curve within the "banana" specification (Table 202-B), the "Design Aggregate Gradation" (DAG), which will be the target for crusher control. The Contractor shall declare the DAG, in writing to the Quality Manager, within production of the first 10% of the Approximate Quantity of a given aggregate classification.

As part of its quality audit program and for payment purposes, the Ministry may sample the stockpile to assess aggregate gradation compliance to contractual requirements.

202.20.02 Variation Limits

The running average of four (4) consecutive tests, on each applicable individual sieve size specified in Table 202-C, shall be maintained at all times within the limits specified in that Table.

The maximum permissible variation from the DAG curve, on each applicable individual sieve size specified in Table 202-C, of the mean of any four (4) consecutive tests shall also be within the limits specified in Table 202-D.